higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

STRENGTH OF MATERIALS AND STRUCTURES N5

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This marking guideline consists of 7 pages.

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QUESTION 1

1.1

$$U_T = U_1 + U_2 = \frac{F^2}{2.E}\left[\frac{L_1}{A_1} + \frac{L_2}{A_2}\right]$$

$$12 = \frac{(19992,78)^2}{2.(215\times10^9)}\left[\frac{0,09}{1,963\times10^{-3}} + \frac{L_2}{3,848\times10^{-3}}\right]$$

$$12909,319 = 45,848 + 259,875.(L_2)$$

$$\therefore L_2 = 49,499 \approx 49,5\ mm$$ ✓

(5)

1.2

$$U_T = \frac{1}{2}.F.X_T$$

$$12 = \frac{1}{2}.(19992,78).X_T$$

$$\therefore X_T = 1,2\times10^{-3}\,m = 1,2\ mm$$ ✓

(3)

1.3

$$\varepsilon_T = \varepsilon_1 + \varepsilon_2 = \frac{F}{E}.\left[\frac{1}{A_1} + \frac{1}{A_2}\right]$$ ✓

$$\varepsilon_T = \frac{19992,78}{215\times10^9}.\left[\frac{1}{1,963\times10^{-3}} + \frac{1}{3,848\times10^{-3}}\right]$$

$$\varepsilon_T \approx 7,154\times10^{-5}$$ ✓

(4)

1.4

$$\sigma = \frac{F}{A_1} = \frac{19992,78}{1,963\times10^{-3}} = 10,185\ MPa$$ ✓

(3)

[15]

QUESTION 2

2.1 2.1.1

$$X_1 = X_2$$

$$\frac{F_1 . L_1}{A_1 . E_1} = \frac{F_2 . L_2}{A_2 . E_2}$$

$$\frac{F_1 . (127)}{1,767 \times 10^{-4} . (215 \times 10^9)} = \frac{F_2 . (127)}{3,142 \times 10^{-4} . (145 \times 10^9)} \checkmark$$

$$F_1 = 0,834 . F_2 \overrightarrow{Eq1}$$

$$F_T = F_1 + F_2$$

$$65000 = 0,834 . F_2 + F_2 \checkmark$$

$$\therefore F_2 = 35444,078\ N \checkmark$$

$$\therefore F_1 = 29555,922\ N \checkmark$$

$$\sigma_1 = \frac{F_1}{A_1} = \frac{29555,922}{1,767 \times 10^{-4}} = 167,266\ MPa \checkmark \checkmark$$

$$\sigma_2 = \frac{F_2}{A_2} = \frac{35444,078}{3,142 \times 10^{-4}} = 112,807\ MPa \checkmark \checkmark$$

(8)

2.1.2 Consider the steel bar:

$$X_1 = \frac{F_1 . L_1}{A_1 . E_1} = \frac{29555,922 . (0,127)}{1,767 \times 10^{-4} . (215 \times 10^9)} \checkmark = 9,88 \times 10^{-5} m \approx 0,099\ mm \checkmark$$

$$L_{Final} = L_{initial} - X_1 = 127 - 0,099 = 126,901\ mm \checkmark$$

(4)

2.2 2.2.1

$$\Delta t.(\alpha_2-\alpha_1).L=F.\left(\frac{1}{A_1.E_1}+\frac{1}{A_2.E_2}\right).L \quad \checkmark$$

$$37.(18\times10^{-6}-12\times10^{-6})=F.\left(\frac{1}{1.767\times10^{-4}.(215\times10^{9})}+\frac{1}{3,142\times10^{-4}.(145\times10^{\text{¿}}}\right) \quad \checkmark$$

$$F=4596,946\ N$$

$$\sigma_1=\frac{F}{A_1}=\frac{4598,946}{1,767\times10^{-4}}=26,027\ MPa \quad \checkmark \quad \checkmark$$

$$\sigma_2=\frac{F}{A_2}=\frac{4598,946}{3,142\times10^{-4}}=14,637\ MPa \quad \checkmark \quad \checkmark$$

$$\sigma_{R1}=\sigma_{1(F)}\pm\sigma_{2(\Delta t)}=-167,266-26,027=193,253\ MPa(tensile) \quad \checkmark \quad \checkmark$$

$$\sigma_{R2}=\sigma_{2(F)}\pm\sigma_{2(\Delta t)}=-172,807+14,637=98,17\ MPa(tensile) \quad \checkmark \quad \checkmark$$

(10)

2.2.2 Consider the steel bar:

$$\delta_{X1}=\alpha_1.\Delta t.L_1-\sigma_1.\frac{L_1}{E_1}=-(12\times10^{-6}).(37).(127)-26,027\times10^{6}.\left(\frac{127}{215\times10^{9}}\right) \quad \checkmark \quad \checkmark$$

$$\text{¿}-0,0718\ mm$$

$$L_{Final}=L_1-X_1-\delta_{X1}=127-0,0988-0,0718 \quad \checkmark$$

$$\text{¿}126,829\ mm \quad \checkmark$$

(4)

[26]

QUESTION 3

Consider the circumferential stress:

$$\sigma_T = \frac{P_i . D}{2 . t . \eta_L}$$

$$150 \times 10^6 = \frac{3 \times 10^6 . (D)}{2 . (0,018) . (0,85)}$$

$$\therefore D = 1,53 \ m$$

Consider the longitudinal stress:

$$\sigma_L = \frac{P_i . D}{4 . t . \eta_C}$$

$$150 \times 10^6 = \frac{3 \times 10^6 . (D)}{4 . (0,018) . (0,52)}$$

$$\therefore D = 1,872 \ m$$

D = 1,53 metres

If D = 1,872 m, the tensile stress developed in the material will be greater than 150 MPa and this will lead to failure of the boiler drum. **[10]**

QUESTION 4

4.1

$$M = \frac{1}{4} . W . L + \frac{1}{8} . \omega . L^2 = \frac{1}{4} . (17) . (1,2) + \frac{1}{8} . (25) . (1,2)^2$$

$$= 5,1 + 4,5$$

$$= 9,6 \ kN/m$$

(3)

4.2

$$\overline{y}_T = \frac{\overline{y_1} . A_1 + \overline{y_2} . A_2}{A_1 + A_2} = \frac{242,6 . (5,876 \times 10^{-3}) + 129,8 . (5,501 \times 10^{-3})}{5,876 \times 10^{-3} + 5,501 \times 10^{-3}}$$

$$= 188,059 \ mm \uparrow$$

(3)

4.3

$$I_{XX_T} = \left[I_{X_1} + A_1 . r_{y_1^2} \right] + \left[I_{X_2} + A_2 . r_{y_2^2} \right]$$

$$= \left[80,28 \times 10^{-6} + 5,876 \times 10^{-3} . (0,055^2) \right] + \left[65,54 \times 10^{-6} + 5,501 \times 10^{-3} . (0,059^2) \right]$$

$$= 1,82 \times 10^{-4} m^4$$

(6)

4.4

$$\sigma_{max} = \frac{M . y_{max}}{I_{XX_T}} = \frac{9600 . (0,188)}{1,82 \times 10^{-4}} = 9,921 \ MPa$$

$$y_{min} = 259,6 - 188,059 = 71,541 \ mm$$

$$\sigma_{min} = \frac{M . y_{min}}{I_{XX_T}} = \frac{9600 . (0,072)}{1,82 \times 10^{-4}} = 3,774 \ MPa$$

(8)

[20]

QUESTION 5

5.1 $J=\frac{\pi}{32}.D^4=\frac{\pi}{32}.(0,06^4)=1,272\times10^{-6}m^4$ ✓

$T=\frac{G.\theta.J}{L}=\frac{85\times10^9.(0,0314).(1,272\times10^{-6})}{1,55}=2190,302\ N.m$ ✓✓

$\tau=\frac{T.R}{J}=\frac{2190,302.(0,03)}{1,272\times10^{-6}}=51,658\ MPa$ ✓✓ (5)

5.2 $P=2\pi.T.\frac{N}{60}=2\pi.(2190,302).\frac{900}{60}=206,431\ kW$ ✓✓ (3)

5.3 $P_{hollow}=1,2.P=1,2.(206,431)=247,717\ kW$ ✓

$P_{hollow}=2\pi.T.\frac{N}{60}$

$247717=2\pi.T.\frac{900}{60}$ ✓

$\therefore T_{hollow}=2628,362\ N.m$ ✓

$\frac{T}{J}=\frac{G.\theta}{L}$

$\frac{2628,362}{J}=\frac{85\times10^9.(0,0314)}{1,55}$ ✓

$\therefore J=1,526\times10^{-6}m^4$ ✓

$J=\frac{\pi}{32}.(D^4-d^4)$

$1,526\times10^{-6}=\frac{\pi}{32}.[D^4-(0,5.D)^4]$

$¿\frac{\pi}{32}.[D^4-0,0625.D^4]$ ✓

$\therefore D=63,811\ mm$ ✓

$d=31,906\ mm$ ✓ (8)

[16]

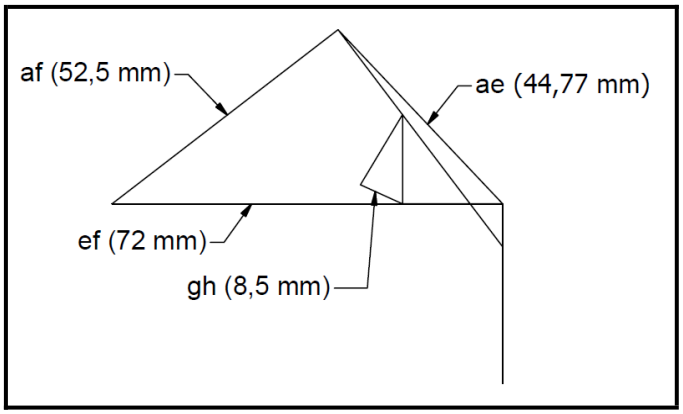
QUESTION 6

6.1 True
6.2 True
6.3 False
6.4 True
6.5 False

(5 × 1) **[5]**

QUESTION 7

$$R_{ROLLER}.(8)=30\sin 45^{\circ}.(4)+25.(6)+25\sin 45^{\circ}.(2)$$
$$\therefore R_{ROLLER(de)}=32,892\ N$$



Vector	Magnitude	Nature
ae	✓ 45 N	✓
af	✓ 52,5 N	S ✓
gh	✓ 8,5 N	T ✓
ef	✓ 72,0 N	T ✓

A tolerance of ± 2 N may be used.

[8]

TOTAL: 100

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